

Zinc/ZnO core-shell hexagonal nanodisk dendrites and their photoluminescence

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Abstract

We report the electrodeposition of novel zinc dendrites composed of self-assembled regular hexagonal zinc nanodisks shelled with ZnO layers. The Zn nanodisks range in diameter from about 100 nm to several hundreds of nanometers and are about 20–40 nm thick. The thickness of ZnO layer is about 3–4 nm. In the as-prepared condition, the photoluminescence (PL) spectra of the dendrites are composed of a violet emission band at about 415 nm and a green emission band at about 550 nm at room temperature. The violet and green emissions are attributed to the radiative recombination of a delocalized electron close to the conduction band with a deeply trapped hole in the V_{Zn}^- and V_{O}^+ centers, respectively. The PL of the dendrites can be tuned by heat treatment. With an increase in the heating temperature, the intensity of the green emission increases, while the intensity of the violet emission decreases.

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1. Introduction

Nanocrystals exhibit a wide range of electrical, magnetic and optical properties, and are therefore of both fundamental and technological interest [1–4]. When developing nanocrystals the most important requirements are shape and size control. Scientific interest is based upon the fact that the shape and size of nanocrystals strongly influence their widely varying properties, and the ability to tune the shape and size of nanocrystals is directly related to the ability to tune their properties [5–10]. To date, many efforts have been devoted to the synthesis of nanocrystalline materials with various shapes, such as nanotubes [11–13], nanowires [14–16], nanorods [17–19], nanobelts [20,21], nanocubes [22] and dendrites [23]. As a new nanostructural morphology,

nanodisks have attracted much attention for their potential applications in information storage media and as nanosensors [24–27]. More recently, Zn/ZnO core-shell nanodisks have been prepared by using a solid-vapor synthetic approach at 1100–1150 °C [28]. This gas-phase method requires high temperature and expensive equipment. Here, we report a simple electrochemical approach for the synthesis of Zinc/ZnO core-shell hexagonal nanodisk dendrites. The morphology control and formation mechanism of the dendrites have been discussed. Finally, the influence of heat treatment on the structure and photoluminescence of the dendrites has been investigated.

2. Experimental section

Electrodeposition of zinc dendrites was performed in a rectangular cell of area 50 mm × 40 mm and height 20 mm. Electrodes were high purity (99.99%) zinc foils covering the centers of the two opposite interior sidewalls

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separated by 50 mm. The cathode was a rectangular foil 10 mm long and 1.5 mm wide, while the anode was a circular foil of diameter 10 mm. A glass plate was placed horizontally just below the cathode to support the deposit which would otherwise break spontaneously during growth due to gravity. The cell was filled with 0.3 M HBO_3 solution as the sole electrolyte up to a level of 2 mm above the glass plate. HBO_3 solution is a weak electrolyte with a high resistance. This choice, instead of other commonly used electrolytes such as H_2SO_4 , allows application of a high electrode voltage without leading to tremendous current and failure of the experiment. Note that the Zn ion concentration in the electrolyte was initially zero. Zn ions dissolved electrochemically into the electrolyte from the anode and were then driven and deposited onto the cathode. Their concentration hence remained very low throughout the whole experiment. This is in sharp contrast to most previous studies in which Zn^{2+} solutions of typical concentration from 0.025 to 1.0 M were used as the electrolyte [29]. The whole cell was kept at 2 °C instead of room temperature commonly adopted. Our results indicate that a lower temperature improves the regularity of the dendrites. Deposition was carried out under a constant cell voltage of 180 V, compared to typically just 2–12 V often used in the literature [29]. Deposition at the cathode was initially extremely slow, but speeded up after about 20 min when considerable amounts of Zn ions had dissolved into the electrolyte. We stopped an experiment before the deposit reached the anode. In order to study the influence of heating temperature, the samples were subsequently heated in air at different temperatures (300–600 °C) for 2 h using a thermal annealing furnace. The structure and morphology of the zinc deposits were investigated using an X-ray diffractometer (*D/Max 2250 V*), a scanning electron microscope (SEM, EM-6300), and a transmission electron microscope (TEM, JEM-2010). Photoluminescence measurements were performed by using a He–Cd laser (325 nm, 10 mW) as the excitation source at room temperature.

3. Results and discussion

Fig. 1 shows the SEM images of the zinc deposits. It can be seen that the zinc deposits consist of small dendrites with regular shapes in random directions (Fig. 1a). We believe that these dendrites constitute not only dangling branches but also the conducting backbone of the overall aggregate, since no other structure of substantial size or connectivity had been identified in these regions. Fig. 1b shows a high-magnification SEM image in which the morphology of individual dendrites can be clearly observed. These dendrites exhibit unique characteristics; they consist of overlapping nanodisks, most of which are hexagonal with very flat facets bounded by sharp edges and corners. The size of these nanodisks and in related experiments ranges from about 100 to 500 nm. They self-assemble into gentle staircases which in turn constitute the main stems

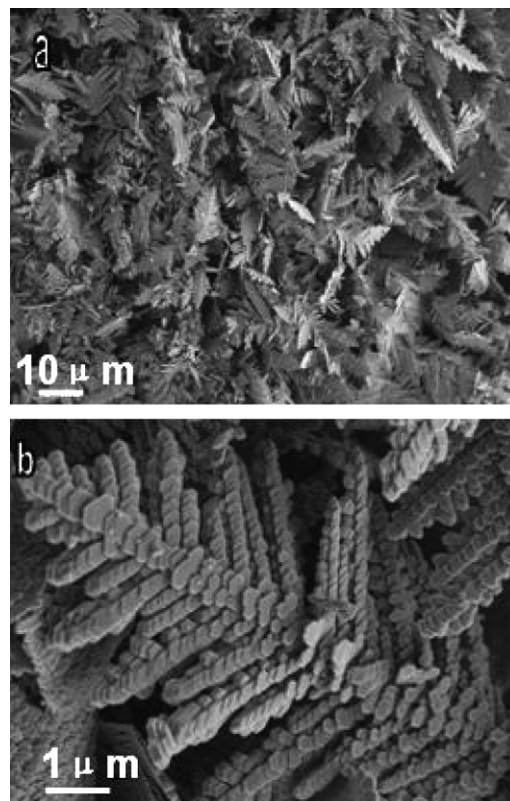


Fig. 1. SEM images of zinc deposits at (a) low and (b) high-magnification.

and side-branches of the dendrites. The side-branches point obliquely forward and subtend an angle of 60° with the main stem as dictated by the hexagonal symmetry of zinc.

To obtain a better understanding of the relation between the morphology and applied voltage, we also performed electrodeposition at other voltages. Fig. 2 shows the SEM images of the deposits prepared at 70 V (a), 90 V (b), 120 V (c) and 180 V (d), respectively. It can be seen that the microstructure of zinc deposits varies sensitively with the voltage. Zinc microrods (Fig. 2a), nanorod dendrites (Fig. 2b), ordinary dendrites (Fig. 2c) and nanodisk dendrites (Fig. 2d) were obtained by adjusting the voltage. Generally, the morphology of a crystal is controlled by both the structure of the crystal and the growth conditions. The anisotropy of growth rates determines the morphology of crystals. The morphology evolution from rods to fine rod dendrites and dendrites with increasing applied voltages resembles the findings in the electrodeposition of lead nanostructures [30], where the formation of nanowires and dendrites requires high negative potential. Preferential growth, including branching growth, is created by a diffusion effect [31]. When a crystal grows, ions near the surface are consumed by the growing crystals and a concentric diffusion field forms around the crystal. This forms the apexes of a polyhedral crystal. The concentration polarization and electrochemical polarization on the apexes are smaller than other parts of the crystal. The apexes protrude further into the region of higher concentration, and grow faster than other parts of the crystal, thus forming rods or needles.

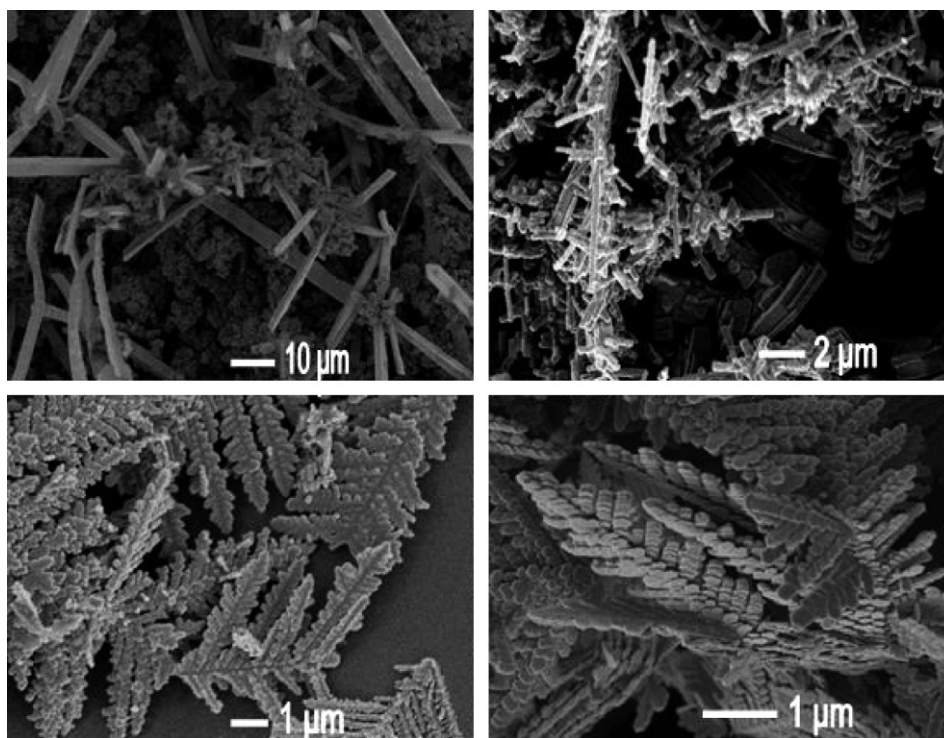


Fig. 2. SEM images of zinc deposits obtained at different voltages: (a) 70 V, (b) 90 V, (c) 120 V, (d) 180 V.

In this case, the applied voltages were much higher than those often used in the literature [29]. The high voltage resulted in high negative potential on the cathode, and the cathode potential becomes more negative when the applied voltage is increased. The more negative potential of the cathode is, the less nucleation energy is required. At higher negative potential, the rod tip is finer and starts to branch. It can be inferred that regular nanodisk dendrites require higher negative potential for periodic nucleation and growth of nanodisks compared to the ordinary dendrites. Their formation mechanism will be discussed later.

The structure of the zinc dendrites have been further investigated by TEM. Fig. 3a and b show such a dendrite and its magnification. According to the TEM observation, the thickness of the nanodisks ranges from 20 to 40 nm. Fig. 3c shows a high-magnification image of a dendrite tip. It can be clearly seen that the dendrite is composed of layered nanodisks. Their corresponding selected-area electron diffraction (SAED) patterns confirm that the nanodisks are single crystalline and the disks are connected along the $\langle \bar{1}2\bar{1}0 \rangle$ orientation (Fig. 3d). In addition to a pure zinc phase, the SAED patterns indicate a second phase which turns out to be ZnO with a Wurtzite-type structure, and the ZnO layer has an epitaxial relationship with the zinc surface. Results from X-ray diffraction (XRD) also indicate the existence of the Wurtzite ZnO phase and will be discussed later. The high-resolution images of the regions indicated in Fig. 3b are shown in Fig. 4. It can be seen that the nanodisks are single crystalline. At the edges of the nanodisks, a thin layer of ZnO is

visible. A ZnO film about 3–4 nm thick completely surrounds the zinc crystallite. The epitaxial relationship between ZnO layer and zinc nanodisk reveals that the ZnO layer is formed from the dehydration of the hydroxide during the deposition process [32]. The zinc dendrites are very sensitive to the electron beam. After irradiation for about 10 s under the electron beam at 200 kV, the zinc sublimated immediately and only ZnO layers were left (Fig. 5). The reaction was found to start at a certain area of the sample, and then the fierce sublimation propagated throughout the dendrites. It is interesting that the remaining ZnO layers retained the morphology of the dendrites, indicating that the zinc cores are completely surrounded by ZnO layers.

Several investigators have reported electrodeposition of zinc dendrites in which the applied potential and electrolyte concentration appear to exhibit a strong influence on the morphology of the zinc deposits [33,34]. However, the mechanism of zinc dendrite nucleation and crystallization at the microscopic scale remains unclear. Here, the novel morphologies of the zinc dendrites should shed light on the formation mechanism. The crystal growth velocity along various planes is dependent upon their atomic packing density. The slowest growth rate is normal to the most densely packed planes. Therefore, the (0001) would have the slowest growth rate, in conformation with the crystal symmetry. Periodic nucleation and growth of neighboring disks is biased towards higher fields, which determines the onset of lateral branches in $\langle \bar{1}2\bar{1}0 \rangle$ directions. The very flat facets, sharp edges and good hexagonal symmetry of the nanodisks indicate that the structure has practically

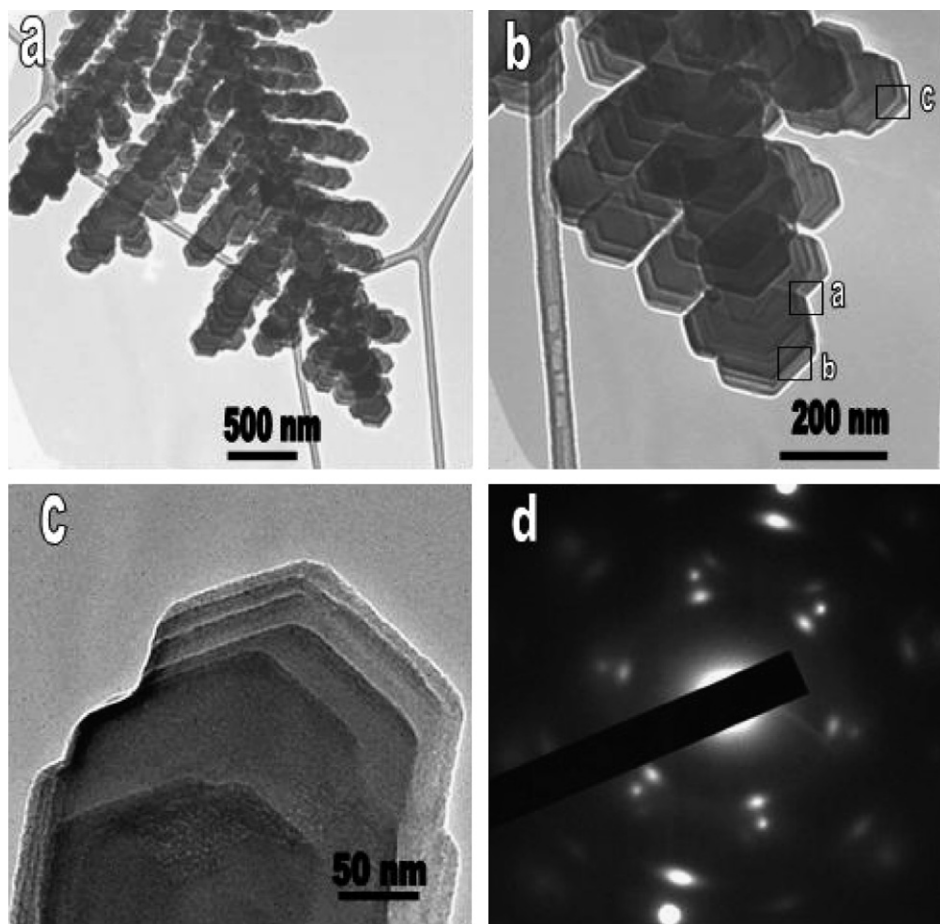


Fig. 3. (a, b) TEM images of a zinc dendrite. (c) The TEM image of the tip of a dendrite, and (d) the corresponding selected-area electron diffraction pattern of the tip in (c).

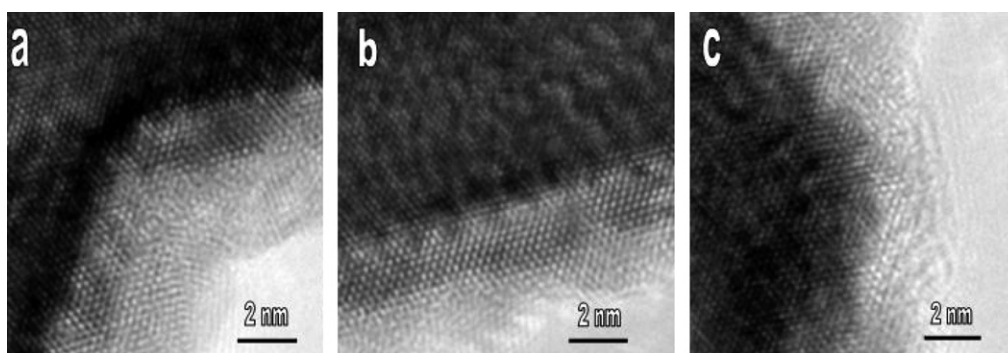


Fig. 4. HRTEM images of the regions indicated in Fig. 3b.

reached local equilibrium for the given thickness. The upper layer nucleation energy barrier on the (0001) plane is therefore significantly larger than other barriers for planar kinetics. Side-branches subtend angles of about 60° with main branches, revealing the hexagonal symmetry of the lattice. More precisely, main stems and side-branches preferentially grow along (1000) or equivalent directions apart from the small tilt in the experiments. Dendritic tips are known to grow towards directions with maximum surface energy [35]. This is consistent with the fact that a

(1000) facet, which is poorly coordinated, has the highest effective surface energy. The (1000) plane of zinc hence should also be of higher energy than, for example, the $(1\bar{1}00)$ facet. With six preferred growth directions, side-branches in general can subtend an angle of 60° or 120° with the main branches, representing respectively obliquely forward and backward geometries. Obliquely forward growth is preferred because the side-branches are then more parallel to the main tips and hence in general point more closely towards the local particle sources.

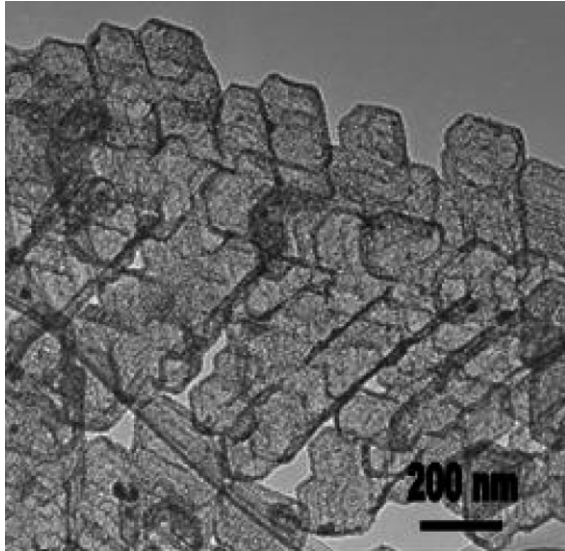


Fig. 5. The TEM image of the remaining ZnO layers after electron beam irradiation.

We investigated the influence of heat treatment on the structure and photoluminescence (PL) of the zinc dendrites. The XRD patterns of the zinc dendrites under different heat treatment conditions are shown in Fig. 6. Intense peaks of hexagonal structure Zn as well as a weak (002) peak of Wurtzite structure ZnO can be observed for the as-deposited zinc dendrites. After heating at 300 °C, the intensities of the ZnO peaks increase and those of the Zn peaks decrease. With a further increase in the temperature, the intensities of the ZnO peaks increase and those of the Zn peaks decrease. After heating at 600 °C, the Zn peaks become negligible, indicating that the Zn cores have been oxidized almost completely. Fig. 7 shows the PL spectra of the Zn dendrites at different heating temperatures. It has been found that temperature has a great effect on the PL of the Zn dendrites. In the as-prepared condition, a violet emission band at about 415 nm and a green emission

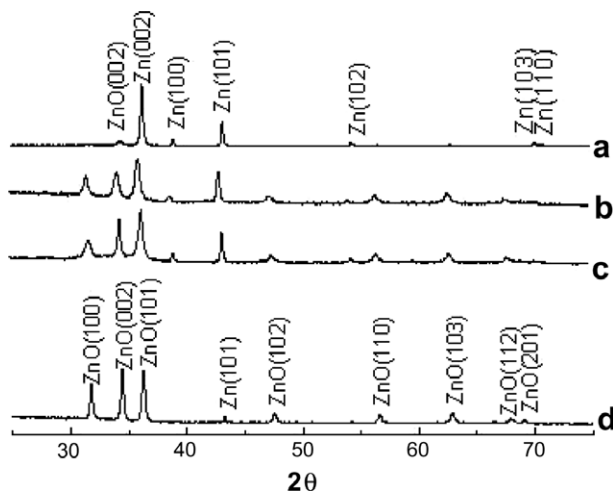


Fig. 6. XRD patterns of the zinc dendrites at different heating temperature: (a) as-deposited, (b) 300 °C, (c) 400 °C, (d) 600 °C.

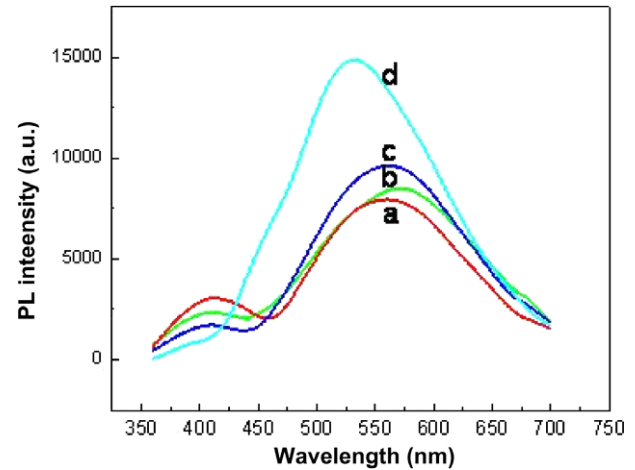


Fig. 7. PL spectra of the zinc dendrites at different heating temperatures: (a) as-deposited, (b) 300 °C, (c) 400 °C, (d) 600 °C.

band at 550 nm can be observed, but the intensity of the violet emission is lower than that of the green emission. After heating at 300 °C, the green emission intensity increases, while the intensity of the violet emission decreases. The ratio of the violet emission intensity to the green emission intensity decreases with the increase in heating temperature. After 600 °C heating, a prominent green emission appears, while the violet emission almost disappears. The mechanism of green emission has been suggested to be mainly due to the existence of various points that can easily form recombination centers [36]. In the present situation, the strong green emission suggests that these ZnO shells should have many point defects, such as oxygen vacancies V_{O}^+ [37,38]. In the annealing process, the Zn core was oxidized at the core–shell interface by the diffused oxygen from the outer surface of the shell. The higher the temperature was, the more ZnO formed. It can be inferred that the ZnO formed during the annealing process features a high density of oxygen vacancies, which enhanced the green emission intensity [38]. The violet emission band has already been observed for ZnO [39–41]. However, the origin of the violet emission of ZnO remains speculative. Chang et al. found that anodically etched porous Zn displayed blue/violet emission at 420 nm. Their analysis suggested that the origin of the blue/violet emission may be related to nonstoichiometric ZnO [39]. Wu et al. investigated the PL and cathodoluminescence (CL) spectra of ZnO films grown on sapphire, and proposed that the violet emission band could be assigned to the V_{Zn}^- centers [40]. Our finding of a decrease in the violet emission intensity with an increase in annealing temperature supports Wu et al.'s proposition. The core–shell Zn/ZnO structure can be considered as a Schottky barrier. Such a metal–semiconductor junction will create an electron depletion region at the interface [42]. During the heating process, the width of the depletion layer will decrease because of the increase in the density of V_{O}^+ . This will lead to a decrease in the density of V_{Zn}^- centers, since V_{Zn}^- centers can only exist in the depletion layer. The violet emission intensity depends on

the density of V_{Zn}^- . Hence, the violet emission intensity decreases with the increase in the annealing temperature. In addition, since V_{O}^+ can exist only in bulk ZnO, the magnitude of the width of the depletion layer directly affects the intensity of the green emission. The thinner the depletion layer, the stronger the green emission intensity. Therefore, a further increase in temperature results in a further increase in the green emission intensity, and a corresponding further decrease in the intensity of the violet emission.

4. Conclusions

In conclusion, using a simple electrochemical synthetic approach, we have successfully fabricated Zn/ZnO core-shell hexagonal nanodisks which aligned to constitute quasi-planar dendrites with high regularity. The zinc core and zinc oxide shell have an epitaxial relationship. These zinc nanodisk dendrites exhibit high sensitivity to electron beam irradiation. The structure and photoluminescence of the Zn/ZnO core-shell nanostructure at different heating temperatures have been studied. The PL spectra are composed of a violet emission band at about 415 nm and a green emission band at about 550 nm at room temperature. The violet and green emissions are attributed to the radiative recombination of a delocalized electron close to the conduction band with a deeply trapped hole in the V_{Zn}^- and V_{O}^+ centers, respectively. With an increase in the heating temperature, the green emission intensity increases and the intensity of violet emission decreases. The metal-semiconductor core-shell structure has some important features: for example, the Zn/ZnO core-shell nanostructure could improve electron transport because the zinc core can act as a current-collecting electrode inside while maintaining the surface area. Considering the large difference between zinc and zinc oxide lattice parameters, the misfit relaxation in the core-shell interface is of great scientific interest [43]. These heterostructured nanodisk dendrites have potential applications in nanotechnology, such as nanotransducers and nanosensors, that can utilize their unique facet-controlled structure. This simple electrochemical synthetic approach may provide a versatile means to fabricate nano- or microcrystals with novel morphologies.

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